

THE EMERGENCE OF COMPLEX BEHAVIOR IN LARGE-SCALE ECOLOGICAL ENVIRONMENTS

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ABSTRACT

We investigate how physical scale and population size govern the emergence of complex behaviors in open-ended ecological environments. In the setting we study, agents receive no explicit reward signal or externally specified learning objective; instead, their policies are updated solely through the population-level dynamics of reproduction, mutation, and selection. Because each action simultaneously modifies the environment state and the composition of the surrounding population, the system constitutes a coupled, non-stationary ecology. Our objective is not to optimize a single high-performance policy, but to characterize how behaviors arise and evolve across large populations under competition and environmental pressure. Leveraging contemporary GPU hardware together with a purpose-built multi-agent simulator, we scale both the environment and the population well beyond the regimes previously reported, sustaining populations exceeding 60,000 concurrent agents, each parameterized by an independently evolved neural network policy. We identify several emergent behaviors—including long-range resource extraction, vision-conditioned foraging, and predation—that arise under competitive and survival pressures. We quantify how the available sensing modalities and the spatial scale of the environment affect the onset of these behaviors, and we find that certain behaviors manifest only above a sufficient threshold in environment size and population, with larger scales measurably increasing the stability and consistency of the resulting dynamics. Although evolutionary computation has an extensive prior literature, our scaling results on modern accelerators indicate concrete new directions for treating ecology as an instrument of machine learning, given the increasing availability of computational resources and efficient simulation frameworks. Experimental code will be released upon acceptance.

1 INTRODUCTION

Since Darwin, the evolutionary emergence of biodiversity and complex behavior has been the subject of study across numerous biological disciplines, including evolutionary theory (Darwin 1859, Simpson 1944), ecology (MacArthur & Wilson 2001), and genetics (Dobzhansky 1937). Formal models of these emergent dynamics span a spectrum from qualitative theories of speciation and adaptive radiation (Coyne & Orr 2004, Schluter 2000) to the quantitative frameworks of population genetics (Fisher 1930, Wright 1931).

Within these fields, decades of mathematical modeling, controlled laboratory experiments, and field observation have substantially advanced our understanding of the mechanisms underlying natural evolution. The empirical study of emergent behavior in situ, however, is constrained by several methodological obstacles. As ecosystems increase in spatial extent and structural complexity, they become correspondingly harder to instrument and to manipulate under controlled conditions. Moreover, where such manipulation is feasible, controlled experimentation at ecosystem scale carries the risk of damaging or displacing the wild populations under study.

To characterize the emergence of complex behavior under competitive and environmental pressure, we study the open-ended evolution of neural network policies within large-scale ecological simulations. In this setting, agents are given neither a task objective nor a reward signal; each agent must instead acquire resources sufficient to survive and reproduce under the environment’s transition dynamics. Policy updates occur exclusively through mutation applied to the parameters an

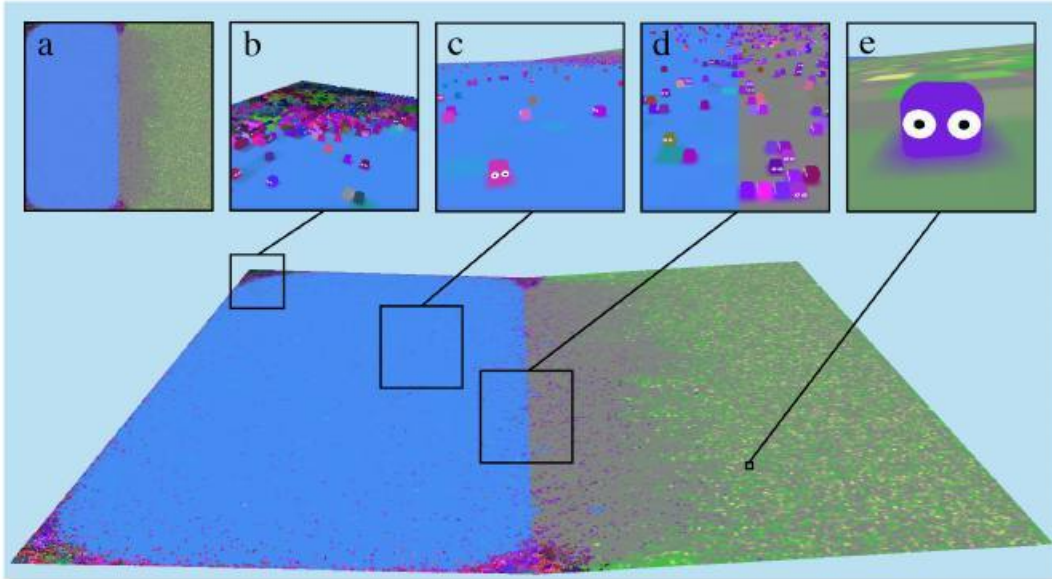


Figure 1: A 3D rendering of a 512×512 map instance from our environment. Inset (a) shows a top-down view of the full map; (b) shows a cluster of agents aggregated in a corner; (c) shows agents occupying water cells; (d) shows agents in transit between land and water; and (e) shows a magnified view of a single agent gathering resources.

agent inherits from its parent. Given the rapid expansion of the computational resources available to researchers, our objective is to study open-ended evolutionary dynamics in populations of these minimal digital organisms at scales not previously attainable, thereby providing new instrumentation for isolating the effects of environment scale and population size on behavioral emergence.

We conduct this study using a new JAX-based simulation environment that supports rapid evaluation of large grid worlds. Our largest configurations comprise over 60,000 concurrent agents within a spatial domain exceeding 1,000,000 grid cells. Agents in this environment must navigate to locate food and other resources in order to survive, and may reproduce once they have accumulated sufficient resources to instantiate a mutated copy of themselves. A 3D rendering of this environment is provided in Figure 1.

Our aim is twofold: to advance the understanding of ecological phenomena through simulation, and to evaluate ecology as a mechanism for producing machine intelligence. Recent work has established that, in machine learning, increased scale does not merely improve model performance incrementally but can qualitatively transform model capabilities (Wei et al., 2022a). Despite this observation, much prior work on embodied learning has been confined to comparatively small environments with limited population sizes. Our work therefore examines the capabilities that emerge when scale is applied not to a single model but to an entire ecosystem. In direct analogy to the observation that certain reasoning capabilities appear in individual models only above a threshold model size (Wei et al., 2022b), we investigate which strategies evolve within populations only once environment size and complexity exceed corresponding thresholds.

Through an extensive suite of experiments, we identify multiple regimes in which the sensor configuration and the spatial scale of the environment jointly determine which behaviors emerge, with consequences that persist over long ecological timescales. Specifically, we show that agents equipped with a simple compass sensor reliably adapt to perform long-range resource-gathering expeditions, whereas this behavior emerges only intermittently at small scales. We further show that agents equipped with a vision sensor evolve more effective foraging and attacking behavior than agents lacking one, and that this advantage likewise manifests more reliably at large scales. Taken together, these experiments demonstrate that the scale and complexity afforded by modern hardware accelerators constitute a practical methodology for studying how evolution in ecological environments gives rise to novel behavior.

2 RELATED WORK

The emergence of life has motivated computational research since the inception of the field. Conway’s Game of Life (Games, 1970) models populations through a set of simple local update rules. Early systems such as Tierra (Ray, 1992) and Avida (Brown,) instantiate life as self-replicating programs subject to mutation. In a similar spirit, Reynolds (1987) demonstrated that local rules of motion are sufficient to induce emergent flocking and schooling behavior.

These principles have been applied to evolve the locomotion morphology (Sims, 1994a,b; Ha, 2018; Gupta et al., 2021; Auerbach et al., 2014; Auerbach & Bongard, 2012; Jeon et al., 2025; Auerbach & Bongard, 2014; Szerlip & Stanley, 2013) and sensing apparatus (Pratt et al., 2022; Mugan & MacIver, 2020; Tiwary et al., 2025) of virtual agents, while neuroevolution methods evolve the neural network architecture itself (Stanley & Miikkulainen, 2002, 2004; Stanley et al., 2009). In contrast to these approaches, we do not seek to produce agents that adapt more effectively to their environment under a reward or objective signal; instead, we aim to characterize the behaviors that emerge under ecological pressure alone and to quantify how environment scale and complexity modulate those behaviors.

Evolutionary principles have likewise informed the internal dynamics of optimization algorithms. Genetic algorithms (Holland, 1992; Jong, 1975) and evolution strategies (Wierstra et al., 2014; Hansen & Ostermeier, 2001; Rechenberg, 1973; Schwefel, 1995; Conti et al., 2018) cast selection and mutation as an optimization procedure. A related line of work employs evolutionary methods explicitly to promote novelty in the resulting solution set (Lehman & Stanley, 2011; Cully et al., 2015). Such methods have further been used to optimize neural network policies for benchmark tasks such as Atari games and MuJoCo locomotion tasks (Salimans et al., 2017; Such et al., 2017; Mania et al., 2018). While our method also employs evolutionary mechanisms to update agent parameters, our objective is to characterize emergent behavior rather than to propose a new optimization algorithm or apply an existing one to a fixed objective.

A further body of work draws on biological inspiration to develop organisms situated within an environment. Early instances include Bedau (1992), Yaeger et al. (1994), Channon & Damber (1998), and Tisue et al. (2004). More recent methods use cellular automata to grow organisms exhibiting emergent morphologies and behaviors (Mordvintsev et al., 2020; Robinson, 2021; Heinemann, 2021; Randazzo & Mordvintsev, 2023; Caussan, 2025). Mordatch & Abbeel (2018) and Park et al. (2023) employ simulated environments to study the emergence of language and communication among agents. Behavioral emergence driven by autotricula (Leibo et al., 2019)—the mechanism by which a population of agents constitutes a progressively more demanding learning landscape for its members—has also been examined, though in populations substantially smaller than ours (Leibo et al., 2018; Team et al., 2021). Complementarily, Wang et al. (2019) and Aki et al. (2024) adopt the inverse formulation, evolving environments optimized for effective agent learning. Although these methods evolve the environment rather than the agent, they inform our work by demonstrating that environmental complexity can elicit sophisticated behavior.

Most closely related to our work are several methods that simulate populations of agents interacting within complex environments. Baker et al. (2019) study a Hide-and-Seek task in which a small set of hider agents learn to exploit environmental tools to evade seeker agents, with both roles trained via reinforcement learning. Related methods train populations against specific task objectives: playing video games (Jaderberg et al., 2019), manipulating objects (Petrenko et al., 2021), or capturing other agents (Zheng et al., 2017; Yamada et al., 2020). Although many of these methods investigate emergence, they do so under fixed tasks and predefined objectives. The Neural MMO framework (Suarez et al., 2019, 2023) adopts a more open-ended reinforcement learning formulation, situating agents within a massively multi-agent game and optimizing for survival rather than a fixed objective. Its population, however, is capped at a fixed number of agents (experiments consider up to 128 concurrent agents) with no generational inheritance, which constrains the analysis of population dynamics. Hamon et al. (2023) is more closely aligned with our setting through its variable population size, genetically inspired evolution, and generational timescale, but its environment is considerably smaller and less feature-rich, and it does not directly investigate the role of scale. Similarly, Lu et al. (2024) simulates agents that evolve to gather resources. That work likewise operates at a substantially smaller scale (a maximum of 256 agents) and emphasizes tool use via programmable

robots. While each of these prior methods advances the study of emergent ecological dynamics, a comprehensive account requires populations that grow and shrink endogenously, evolve without predefined objectives, and interact within environments rich enough to support diverse strategies at scales at which collective patterns become observable. Our work integrates these requirements, enabling analysis of ecological phenomena inaccessible in prior work.

Finally, our work is informed by recent systems that prioritize simulation speed and scale, using optimized libraries to increase throughput and the number of agents that can interact within game-like environments (Koyamada et al., 2023; Matthews et al., 2024; Lechner et al., 2023). We build directly on this progress in computational efficiency, adopting comparable JAX-based acceleration techniques to achieve large scale and high simulation throughput in ecological environments.

3 METHOD

3.1 ECOLOGICAL GAMES

We formalize the interaction between a population of agents and its environment as a class of multiplayer game that we term an **Ecological Game (EG)**. In analogy to the Markov decision process (MDP) in reinforcement learning and the stochastic game (SG) in multi-agent reinforcement learning (see Sutton et al. (1998) and Albrecht et al. (2024)), an EG possesses Markovian transition dynamics but is constructed to represent a time-varying population of agents in the absence of any objective. As in an SG, an EG is specified by a set of agents I , a state space S , an initial-state distribution ρ_0 , an action space A , and a transition function $\mathcal{T}$ that maps a state s_t together with a per-agent action profile $a_{t,i}$ to a distribution over successor states s_{t+1} . In contrast to an SG, an EG omits the reward function and instead introduces a population function $\Gamma(s_t)$ that reports, at each time t , which agents in I are currently alive together with the identity of their parents, so that birth and death events can be inferred from the state trajectory. Just as MDPs and SGs admit partially observable counterparts—POMDPs and POSGs—an EG admits a partially observable variant, which we denote a POEG. A POEG differs from an EG in that no agent observes the full state; the state is instead mediated by an observation function $O(s_t, i)$ that maps a state and an agent index to that agent’s observation. Our environment is a POEG whose observation function is determined by the sensor configuration specified in the following subsection.

3.2 ENVIRONMENT

The specific POEG instantiated in our experiments is a large grid world populated with resources that agents must collect in order to reproduce and survive. The environment maintains a factored state space $s_t = \{s_t^m, s_t^a\} \in S = S^m \times S^a$, in which s_t^m denotes the 2D map-level fields—terrain height and the spatial distribution of resources—and s_t^a denotes the per-agent state, comprising each agent’s position, health, and internal resource stores.

Terrain: Each map is defined by a rock layer that assigns an elevation to every grid cell. Water is initialized in all cells whose rock elevation lies below a fixed sea level. Over the course of an episode, water is redistributed by a discrete flow simulation that transports it downhill according to the elevations specified by the rock layer. In the experiments below we employ six global terrain configurations, shown in Figure 2: **Ocean**, **Beach**, **Island**, **Lake**, **Isthmus**, and **Channel**. We run experiments at grid resolutions of 64×64 , 128×128 , 256×256 , 512×512 , and 1024×1024 . When the grid resolution is varied, the initial population size is scaled proportionally to the change in total grid area. After initialization, the population size evolves endogenously according to the environment dynamics.

Resources and Agent Health: To survive and reproduce, agents must collect three resource types: **water**, **energy**, and **biomass**. Biomass and energy are initialized with a random spatial distribution and are thereafter consumed and redistributed through agent activity.

Every action incurs a small fixed cost in energy and water. Expended water is returned to the landscape, where it flows downhill toward the nearest body of water, whereas expended energy is destroyed. Biomass functions as the reproductive resource: each agent requires a fixed biomass endowment simply to persist, and thus cannot reproduce until it has accumulated excess biomass sufficient to endow an offspring. As with water, the total biomass in the environment is conserved.

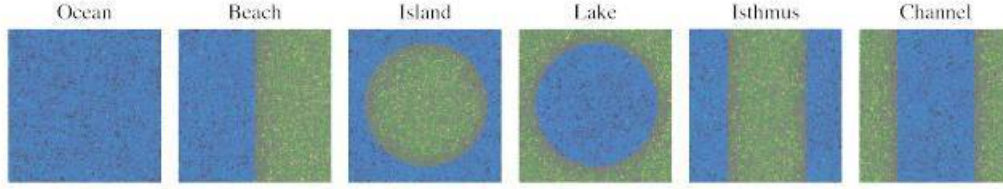


Figure 2: Overhead renderings of the terrain configurations used in our experiments at 256×256 grid resolution. Small dark markers denote individual agents; green and yellow denote spatial concentrations of energy and biomass, respectively. Each image was captured at step 5000 of an episode, after most aquatic biomass has been consumed but before agents have evolved to sustain prolonged survival on land.

over the simulation; unlike water, however, biomass is stationary and does not flow. Energy, in contrast, is not conserved but is generated at a slow rate on free-standing biomass. These resource dynamics were designed to pose a strictly harder acquisition problem than the collection of a single food source. Ablations on the resource rules are reported in Appendix F.2

Each agent additionally maintains a health-point (HP) budget that must remain positive to avoid death. Overspending a critical resource induces rapid HP decay, while sustained energy and water reserves permit slow HP recovery. An agent may also attack other agents, decrementing the targets' HP. In addition, each agent incurs a small, monotonically increasing damage rate as a function of its age, so that agents eventually die of senescence in the absence of other causes. When an agent's HP reaches zero, it dies and all resources it was carrying are deposited into the grid cell it occupied. Consequently, an agent that attacks and kills another agent may subsequently consume the deposited resources, constituting a rudimentary form of predation. Ablations on the HP hyperparameters are reported in Appendix F.8

Sensors: The observation function is composed from four sensors. The **internal** sensor reports the agent's own age, health, and carried-resource quantities; the **external** sensor reports the resources co-located in the agent's current grid cell; the **compass**, motivated by the magnetoreception observed in birds and other animals, encodes the agent's global heading as a 4-way one-hot vector; and **vision** supplies a 7×7 top-down rendered patch centered on the agent, comprising three color channels together with a fourth channel encoding the local elevation gradient. Ablations over the vision configuration are reported in Appendix F.6

Actions: The action space A is finite and discrete, partitioned into five categories. The **Rest** action performs no state change and incurs the minimum energy cost. The three **Move** actions are: rotate clockwise, rotate counterclockwise, and advance one grid cell in the current heading. Cell occupancy is exclusive: if an agent attempts to move into a cell occupied by another agent, or if two agents attempt to move into the same cell, the conflicting move actions are canceled. A single **Attack** action eliminates all agents within the 3×3 neighborhood immediately ahead of the acting agent. Ablations on the attack strength are reported in Appendix F.7. Three distinct **Eat** actions, one per resource type, each transfer a fixed quantity of the corresponding co-located resource from the environment into the agent's stomach, which has a fixed capacity.

Finally, a single **Reproduce** action instantiates an offspring via single-parent reproduction. The action succeeds only if the agent has accumulated resources sufficient to endow the offspring. The offspring is instantiated in the cell directly behind the parent, and its policy is a mutated copy of the parent's parameters. The offspring additionally inherits a fixed quantity of the parent's resources at instantiation.

In the experiments that follow, we selectively disable individual sensors or the attack action in order to isolate the ecological effect of agents with differing capabilities. We consider three sensing configurations: internal and external resource sensors only (**R**); internal, external, and compass (**RC**); and internal, external, compass, and vision (**RCV**). The suffix **+A** indicates that attacking is enabled; for example, (**RC+A**) denotes an agent with resource sensors, a compass, and the attack action.

3.3 POLICIES

Each agent policy is a small, memoryless multilayer perceptron (MLP). We adopt a memoryless architecture to keep the computational model minimal and to reduce the parameter count. The raw output of each sensor is flattened, normalized to the interval $[-1, 1]$, and passed through a dedicated linear layer with output dimension 64. The resulting per-sensor embeddings are summed and passed through a 2-layer MLP with hidden dimension 64 and ReLU activations. Because A is discrete, the output layer is linear with dimension equal to the cardinality of the action set. Actions are sampled from the softmax distribution over these logits, with a temperature parameter that is itself heritable and evolves jointly with the network weights. All weights are stored in half-precision bfloat16 to accommodate larger populations and reduce wall-clock runtime. The per-agent parameter count ranges from approximately 10k to 25k, depending on the active sensor set. Ablations on network size and architecture are reported in Appendix F.5.

Across all runs, the initial population’s weights are drawn using Kaiming initialization (He et al., 2015) and all bias vectors are initialized to zero. The policy of an agent born during an episode is obtained by copying the parent’s weights and adding independent Gaussian noise with standard deviation 3×10^{-2} to every mutable weight. This mutation operator follows prior neuroevolution work (Such et al., 2017). Although more elaborate operators such as crossover are available, we deliberately restrict reproduction to this single mechanism in order to study the effect of environment scale in the simplest possible setting. Empirically, a comparatively large mutation standard deviation was required as a consequence of the low-precision weight representation. An ablation over the mutation rate is reported in Appendix F.4.

Each agent additionally carries a 3-channel color trait that governs its rendered appearance on the map. This trait mutates under the same operator as the network weights and determines the agent’s appearance as observed by other agents through the vision sensor. In principle, this channel provides a substrate by which agents may discriminate among distinct classes of conspecifics.

3.4 COMPUTATION AND REPRODUCIBILITY

All experiments are implemented in JAX (Bradbury et al., 2018) and executed on Nvidia H100 and A40 GPUs within a state-of-the-art simulation environment. Unless otherwise noted, each run is executed for 2M environment steps or until population extinction, whichever occurs first; per-seed wall-clock times range from under one hour at the 64×64 resolution to approximately 10 hours at the 1024×1024 resolution on a single H100 GPU. Appendix E details the simulator architecture, reports additional scaling measurements, and discusses reproducibility considerations.

4 EXPERIMENTS

In the experiments that follow, we quantify the effects of the sensing modality, the environment scale, and the population size on the emergent behavior of agents.

4.1 LONG-DISTANCE RESOURCE GATHERING

Our first experiment characterizes the emergent capacity of agents to traverse long distances inland to acquire resources. We show that, in environments with pronounced directional structure, agents equipped with a compass (**RC**) consistently develop substantially more effective long-range resource-gathering behavior than a baseline agent class (**R**) lacking a compass. We further show that the emergence of this behavior is both more consistent and more effective at larger environment scales.

We first analyze a representative trajectory of the (**R**) agent class, shown in Figure 3. By step 5000 (a), all agents initialized on dry land have died from water depletion, while those in the water have proliferated rapidly owing to the abundant initial resources. At this early stage, the population has not yet evolved policies capable of efficiently locating resources. From this initial population peak, agents in the water intermittently wander onto land (b); once there, they typically fail to return to the ocean and die, depositing their biomass on land. By step 50,000 (c), this incremental transfer has nearly doubled the quantity of uneaten biomass on land. Because this process depletes the aquatic

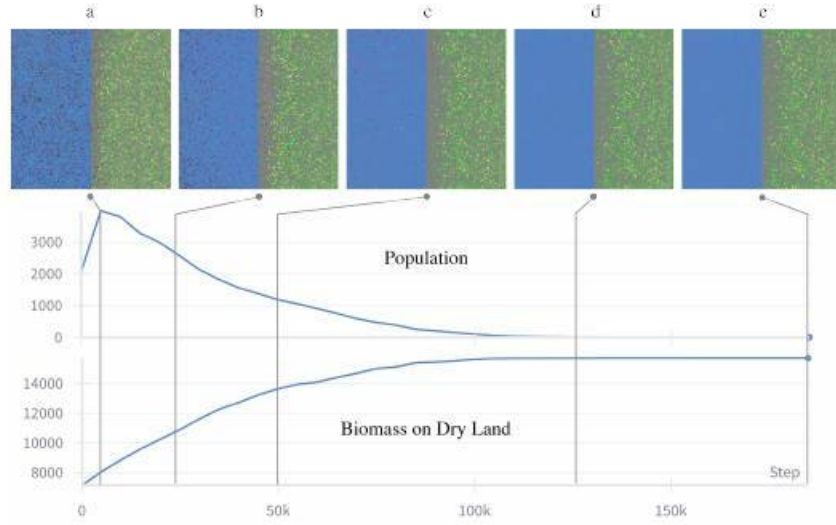


Figure 3: A representative trial with **(R)** agents in a 256×256 **Beach** environment. The top panel plots population size against simulation time; the bottom panel plots the free biomass on dry land.

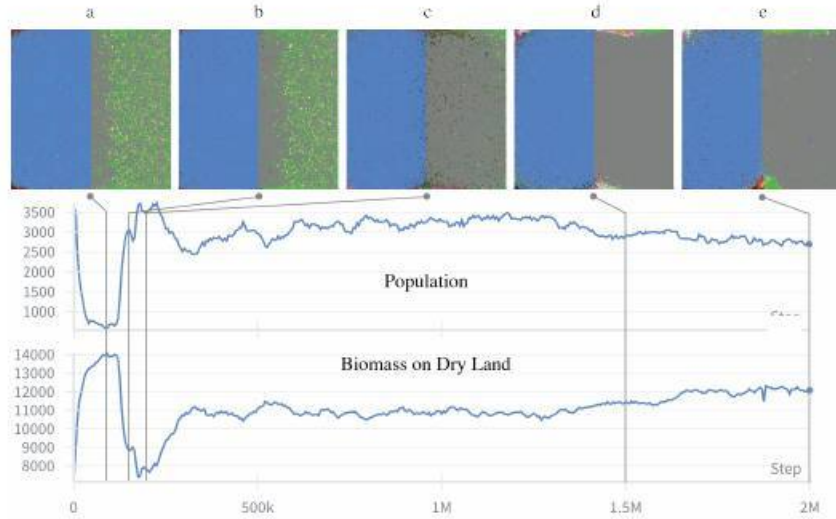


Figure 4: A representative trial with **(RC)** agents in a 256×256 **Beach** environment. The top panel plots population size against simulation time; the bottom panel plots the free biomass on dry land.

biomass below the level required to sustain a large population, the population contracts. By step 125,000 (d), nearly all biomass has been transferred to land and the population is close to extinction. The final agent dies at approximately step 185,000 (e).

This trajectory illustrates the ecological consequence of the agents' inability to return to water. Beyond the death of individuals, the deaths themselves reconfigure the spatial distribution of resources in ways that materially affect the entire population. Simultaneously, this ecological collapse creates an exploitable opportunity for any agent that evolves a policy able to traverse onto land, forage the abundant biomass deposited by dead conspecifics, and return safely to water.

In the next trajectory, shown in Figure 4 the agents are additionally equipped with a compass **(RC)**. The initial phase (a) reproduces the previous pattern: the population contracts while the free biomass on land increases rapidly. In this case, however, a subset of agents exploit the compass to drift toward the relative safety of the corners. By approximately step 150,000 (b), some agents begin to "mine"

the biomass on land, using the compass to navigate repeatedly between land and water. By step 200,000 (c), these miners have cleared the entire beach. Thereafter the population exhibits only minor fluctuations, with agents forming clusters along the edges (d), and remains largely stable through the full 2,000,000 steps (e).

Table 1: Mining events in the **Beach** world for populations of agents with a compass (**RC**) versus populations without a compass (**R**), across five world sizes and corresponding initial populations.  denotes the number of seeds exhibiting a mining event.  denotes the number of seeds in which the population went extinct before 2M steps. Per-seed detail is provided in Table 3 in Appendix A

	1024 × 1024		512 × 512		256 × 256		128 × 128		64 × 64	
	32768		8192		2048		512		128	
										
(RC)	4/4	0/4	4/4	0/4	4/4	0/4	3/4	1/4	2/4	3/4
(R)	1/4	0/4	0/4	4/4	0/4	4/4	0/4	4/4	0/4	4/4

We quantify these observations with a full factorial set of **Beach** experiments spanning grid resolutions 1024×1024 , 512×512 , 256×256 , 128×128 , and 64×64 , with corresponding initial populations of 32768, 8192, 2048, 512, and 128 agents. Each configuration is run with four random seeds, and for each run we record whether mining behavior emerges, operationalized as a decrease of at least 10% in the free biomass on dry land relative to its historical maximum. For each mining event we additionally record its onset time, defined as the time of the local maximum in land biomass, together with the percentage decrease in biomass. Aggregate mining and extinction outcomes are summarized in Table 1 with per-seed detail provided in Table 3 in Appendix A. Mining behavior emerges consistently in (**RC**) agents but rarely in (**R**) agents. The consistency of mining increases with environment scale: it occurs in all four seeds at resolutions of 256×256 and larger. At smaller scales, populations are less stable and exhibit a higher probability of early extinction.

To characterize these effects more precisely, we run additional seeds with (**R**) and (**RC**) agents at the 512×512 resolution (32 seeds) and the 128×128 resolution (64 seeds) and apply a Kaplan–Meier estimator to the probability of extinction and of mining as a function of simulation time. We find that extinction is rarer and slower to occur at larger world sizes, whereas mining is more frequent and occurs earlier at smaller grid sizes. We additionally fit a Cox proportional-hazards model to the timing of mining discovery and of extinction as a function of world size and compass availability; consistent with the non-parametric analysis, larger world size substantially reduces the extinction hazard and markedly accelerates mining discovery relative to smaller worlds, and enabling the compass produces an effect of comparable magnitude. Full details are provided in Appendix B

Table 2: Mining and extinction events in the **Island**, **Lake**, **Isthmus**, and **Channel** terrains for populations of agents with a compass (**RC**) versus those without a compass (**R**), at the 512×512 resolution.  denotes the number of seeds exhibiting a mining event.  denotes the number of seeds in which the population went extinct before 2M steps. Per-seed detail is provided in Table 4 in Appendix A

	Island		Lake		Isthmus		Channel	
								
(RC)	4/4	0/4	2/4	2/4	4/4	0/4	3/4	1/4
(R)	3/4	1/4	2/4	4/4	1/4	3/4	0/4	4/4

We run analogous experiments in the **Island**, **Lake**, **Isthmus**, and **Channel** terrains at the 512×512 resolution with (**R**) and (**RC**) agents in order to isolate the effect of terrain configuration on the emergence of mining behavior. Aggregate mining and extinction outcomes for these experiments are reported in Table 2 with per-run detail in Table 4 in Appendix A. Consistent with the **Beach** results, equipping agents with a compass yields earlier and more consistent emergence of mining as well as more stable populations with lower extinction rates. (**RC**) agents adapt more readily to

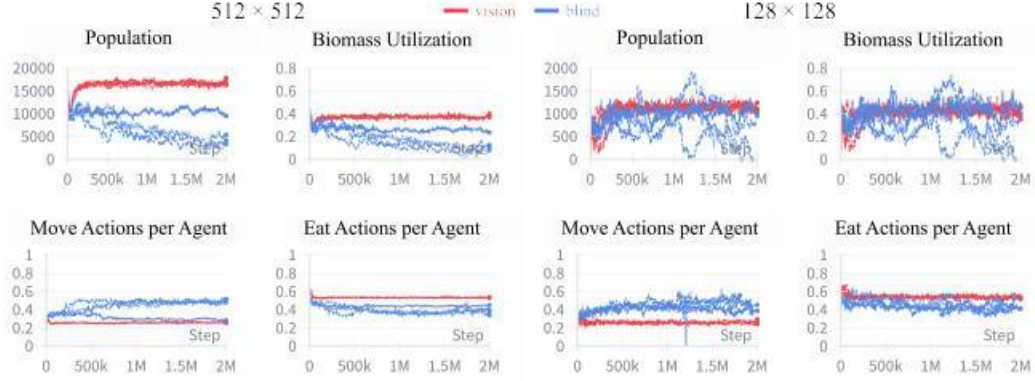


Figure 5: Population size, biomass utilization, move actions per agent, and eat actions per agent in 512×512 and 128×128 **Ocean** worlds. Biomass utilization, move actions per agent, and eat actions per agent are averaged over agents at each time step. Each panel plots four seeds with vision agents (**RCV + A**) (red) and four seeds with blind agents (**RC + A**) (blue). With vision, populations are larger and more stable and achieve greater biomass utilization, attributable to reduced movement and more frequent, more effective eating. At larger scales these dynamics are more stable, with lower inter-run variance, indicating that the marginal advantage of vision over blindness for adapting foraging behavior increases with environment scale. Figure 9 in Appendix C provides the corresponding comparison for the 1024×1024 and 256×256 resolutions.

the island and isthmus terrains than to the lake and channel, in which they occasionally go extinct. We attribute this to the availability of a simple heading heuristic: in the island and isthmus terrains, moving due east or due west always returns an agent to water, whereas no comparably simple rule exists for the channel and lake. The (**R**) agent fails to reliably adapt to any of these terrains, but is least impaired on the island. Figure 12 in Appendix D provides a visualization of mining in the island terrain.

4.2 VISUAL FORAGING AND PREDATION

We next conduct experiments in the **Ocean** environment to determine whether agents can adapt to use the vision sensor to locate free resources or to prey on other agents. We compare (**RCV+A**) agents, equipped with resource sensing, a compass, and a 7×7 vision sensor, against (**RC+A**) agents, equipped only with resource and compass sensors.

Figure 5 reports population size, biomass utilization, move-action frequency, and eat-action frequency for agents at the 512×512 and 128×128 resolutions. Corresponding plots for the remaining resolutions are provided in Figure 9 in Appendix C and summary statistics for these plots are reported in Table 5 in Appendix A. Biomass utilization is defined as the fraction of the total environmental biomass currently held within agents. In episodes in which agents possess vision (red curves), agents allocate a smaller fraction of actions to movement and a larger fraction to eating. These runs additionally sustain larger populations and achieve higher biomass utilization. Taken together, these dynamics indicate that the agents have adapted their policies to exploit the vision sensor for more efficient foraging. Across environment scales, the inter-run variance decreases substantially at larger scales. Notably, runs equipped with the vision sensor exhibit markedly more stable dynamics, with reduced temporal variation across all reported metrics.

Figure 6 reports the mean attacks per agent and homicides per attack for these runs. Agents with vision attack relatively infrequently while maintaining a kill rate of approximately 75%. At large grid resolutions, agents without vision occasionally also attain high kill rates, but with substantially less stable dynamics. The underlying cause of this latter effect is difficult to isolate owing to confounding factors. Qualitative inspection (Figure 13 in Appendix D) indicates that these elevated kill rates sometimes coincide with the formation of dense agent clusters at a single location, and that such clusters almost never form in runs with vision agents. This observation is also consistent with vision agents adapting to avoid one another. At smaller scales, the difference between vision and blind agents is less pronounced, but vision-agent populations remain considerably more stable.

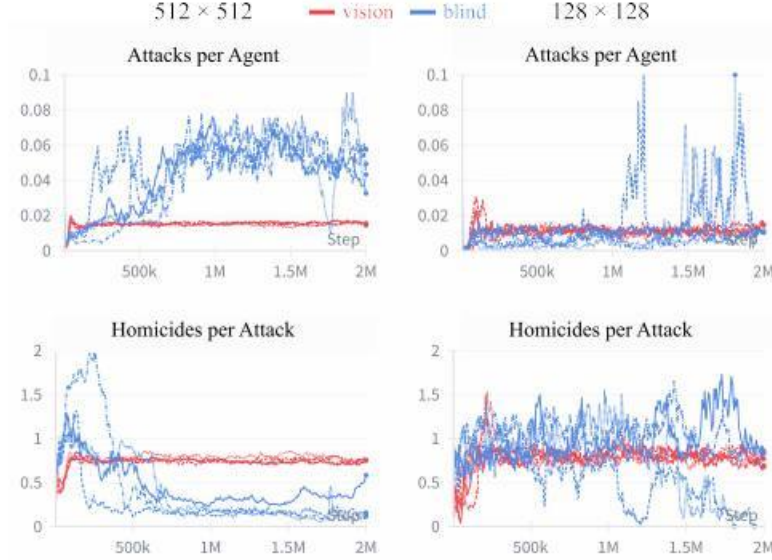


Figure 6: Attacks per agent and homicides per attack, averaged over agents, in 512×512 and 128×128 **Ocean** worlds. Each panel plots four seeds with vision agents (**RCV + A**) (red) and four seeds with blind agents (**RC + A**) (blue). With vision, agents attack less frequently but with substantially higher precision, consistent with the emergence of specialized predation behavior. This effect is more pronounced at larger scales. Figure 10 in Appendix C provides the corresponding comparison for the 1024×1024 and 256×256 resolutions.

4.3 ABLATIONS

In Appendix F we report ablations over individual hyperparameters—initial population size (F.1), resource rules (F.2), mutation rate (F.4), agent network configuration (F.5), vision range (F.6), attack strength (F.7), and agent HP (F.8)—in order to quantify the sensitivity of the emergent behaviors to the simulator configuration. In addition, the experiments in Appendix E.3 verify that the dynamics observed at larger scales are not attributable to statistical self-averaging.

5 CONCLUSION

This study establishes open-ended ecological settings as a viable paradigm for investigating the emergence of intelligence. Across experiments in worlds of varying scale and complexity, we find that both environment scale and sensory richness contribute substantially to the development of sophisticated sensing and decision-making policies. These results indicate that ecological and evolutionary pressure, coupled with a sufficiently rich environment, constitutes a plausible pathway toward the development of increasingly capable artificial agents. Our experiments further demonstrate that such behaviors can be evolved in large populations within a modest hardware budget and in the absence of any explicit objective. We anticipate that these results will support continued investigation of ecology as a mechanism for both developing and analyzing artificial intelligence.

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A ADDITIONAL DATA

Table 3: Mining events in the **Beach** world for populations of agents with a compass (**RC**) versus populations without a compass (**R**), across five scales.  indicates whether the run exhibited a mining event.  denotes the percentage decrease in free biomass on dry land over the mining event, and  denotes the approximate timestep at which the free dry biomass attained the local maximum marking the onset of the mining event. For runs with multiple mining events, we report the first.  denotes the timestep of extinction, or n/a for runs that persisted through 2M steps.

	1024x1024 32768				512x512 8192				256x256 2048				128x128 512				64x64 128			
Compass On (RC)																				
Seed																				
0	✓	35k	11.1	n/a	✓	105k	33.	n/a	✓	145k	49.3	n/a	✓	105k	72.5	n/a	x			
1	✓	90k	16.2	n/a	✓	50k	17.2	n/a	✓	50k	46.6	n/a	x		140	✓	70k 85.3 675			
2	✓	40k	15.	n/a	✓	30k	14.3	n/a	✓	55k	34.7	n/a	✓	50k	50.7	n/a	✓ 10k 78. n/a			
3	✓	60k	15.	n/a	✓	45k	15.6	n/a	✓	90k	47.4	n/a	✓	20k	52.	n/a	x 135			
Avg	4/4	34k	14.3	0/4	4/4	58k	20.	0/4	4/4	85k	44.5	0/4	3/4	58k	58.4	1/4	2/4 40k 81.7 3/4			
Compass Off (R)																				
Seed																				
0	x			n/a	x			945k	x			185k	x			65k	x 45k			
1	✓	275k	12.4	n/a	x			1085k	x			110k	x			55k	x 40			
2	x			n/a	x			1070k	x			165k	x			55k	x 25			
3	x			n/a	x			280k	x			125k	x			90k	x 40			
Avg	1/4	275k	12.4	0/4	0/4			n/a	0/4			4/4	0/4			4/4	0/4 4/4			

Table 4: Mining events in the **Island**, **Lake**, **Isthmus**, and **Channel** terrains for (RC) agents versus (R) agents. The grid resolution is 512×512 with an initial population of 8192.

Island					Lake				Isthmus				Channel			
Compass On (RC)																
Seed																
0	✓	40k	94.3	n/a	x			45k	✓	45k	33.	n/a	✓	50k	27.4	n/a
1	✓	30k	94.3	n/a	✓	30k	22.5	n/a	✓	55k	37.3	n/a	✓	80k	30.8	n/a
2	✓	30k	94.8	n/a	x			45k	✓	40k	28.1	n/a	✓	50k	41.	n/a
3	✓	70k	95.1	n/a	✓	35k	25.3	n/a	✓	30k	41.5	n/a	x			80k
Avg	4/4	43k	94.6	0/4	2/4	33k	23.9	2/4	4/4	43k	35.	0/4	3/4	60k	33.1	1/4
Compass Off (R)																
Seed																
0	x			185k	x			125k	x			145k	x			100k
1	✓	75k	30.9	n/a	x			130k	x			260k	x			105k
2	✓	85k	47.	n/a	✓	75k	23.4	1400k	✓	95k	18.8	n/a	x			105k
3	✓	115k	39.5	n/a	✓	70k	24.3	1250k	x			140k	x			100k
Avg	3/4	92k	39.1	1/4	2/4	73k	23.9	4/4	1/4	95k	18.8	3/4	0/4			4/4

Table 5: Population and foraging measurements for Vision and Blind agents in the **Ocean** world across scales. denotes population size. denotes biomass utilization, the fraction of the total system biomass currently held within the agents' stomachs. denotes move actions per agent, the fraction of steps on which an agent took a move action, averaged over agents. Analogously, denotes the fraction of agent actions that were eat actions. Reported values of , , , and are averaged over all steps of the seed.

	1024x1024 32768				512x512 8192				256x256 2048				128x128 512			
Vision (RCV+A)																
Seed																
0	58354	.396	.402	.426	16413	.369	.257	.532	4195	.382	.255	.536	1126	.426	.261	.541
1	65107	.365	.257	.529	16200	.364	.256	.533	4142	.377	.258	.536	1103	.415	.262	.537
2	64735	.363	.257	.530	16560	.372	.257	.533	4269	.388	.260	.536	1041	.392	.254	.542
3	64595	.362	.256	.532	16206	.365	.255	.531	4249	.388	.259	.533	1100	.414	.257	.538
Avg	63198	.371	.293	.504	16345	.368	.256	.532	4213	.384	.258	.535	1092	.412	.259	.539
Blind (RC+A)																
Seed																
0	22734	.143	.446	.372	6549	.173	.442	.400	2036	.209	.433	.461	545	.219	.465	.427
1	23670	.151	.426	.388	6939	.180	.438	.396	3107	.321	.390	.416	1082	.444	.416	.460
2	24745	.159	.358	.426	5448	.136	.474	.374	1587	.163	.440	.433	979	.421	.404	.436
3	20004	.130	.469	.359	10354	.265	.311	.439	3164	.324	.321	.450	788	.317	.431	.477
Avg	22788	.146	.425	.386	7323	.189	.416	.402	2473	.254	.396	.440	848	.351	.429	.450

Table 6: Predation measurements for Vision and Blind agents in the **Ocean** world across scales. 🗡️ denotes attacks per agent, the fraction of steps on which an agent took the attack action, averaged over all agents. 🎯 denotes homicides per attack, the number of agents killed per attack action. Reported values of 🗡️ and 🎯 are averaged over all steps of the seed. The value 0.562 for 🗡️ in seed 3 of the 128×128 (**RC+A**) run is an outlier: this seed went extinct before 2M steps, producing a spike in the attacks-per-agent metric as the population dwindled.

	1024x1024		512x512		256x256		128x128	
	32768		8192		2048		512	
Vision (RCV+A)								
Seed								
0	.011	.776	.015	.731	.013	.784	.010	.788
1	.016	.733	.015	.741	.014	.780	.011	.793
2	.016	.707	.015	.737	.014	.750	.012	.771
3	.016	.712	.015	.780	.013	.759	.011	.765
Avg	.015	.732	.015	.747	.014	.758	.011	.779
Blind (RC+A)								
Seed								
0	.070	.218	.042	.437	.005	1.074	.016	.558
1	.054	.300	.044	.383	.016	.871	.006	1.032
2	.076	.236	.048	.238	.031	.383	.010	.979
3	.059	.203	.044	.456	.033	.446	.562	.841
Avg	.065	.239	.045	.378	.021	.693	.148	.853

B DEMOGRAPHY

We compare the rates of extinction and of mining emergence as functions of world size and of agent access to global heading via the compass sensor. Extinction is defined as the population reaching zero individuals, and mining behavior is defined as in the main experiments (a 10% decrease in free biomass on land).

Runs are stratified jointly by world size and compass access. We simulated 64 runs each of 128×128 without compass and 128×128 with compass, and 32 runs each of 512×512 without compass and 512×512 with compass. For each of the four strata, we fit a non-parametric Kaplan–Meier estimator to the time-to-extinction distribution, treating runs that did not go extinct within 2M steps as right-censored. As shown in Figures 7 and 8 worlds tend to survive longer and to develop mining more rapidly when they are larger or when agents have access to compass information.

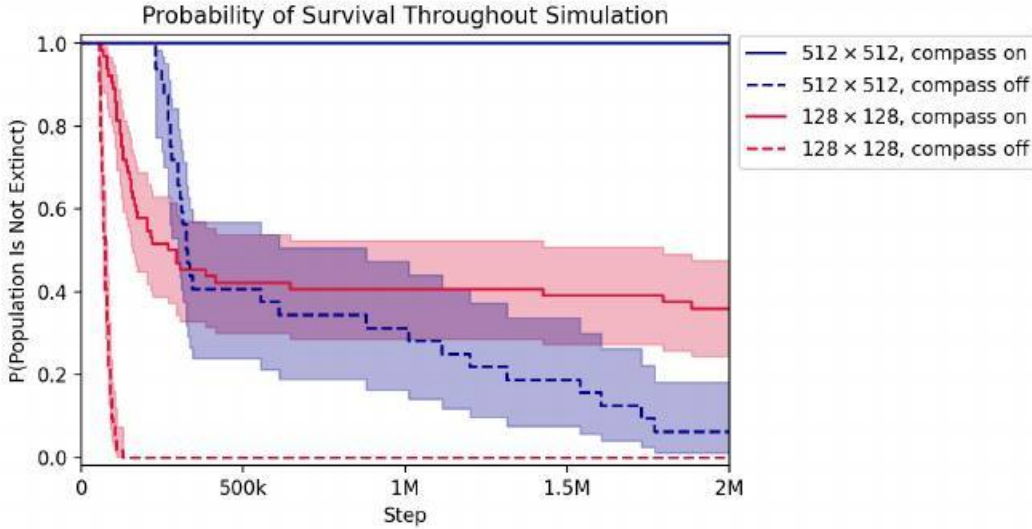


Figure 7: Estimated probability that the population has not gone extinct, conditioned on world size, compass access, and timestep. A separate non-parametric Kaplan–Meier estimator is fit for each stratum of runs (e.g., 512×512 , compass on). Shaded regions denote the 95% confidence interval for each curve. Each 512×512 stratum is fit on 32 runs and each 128×128 stratum on 64 runs.

We further compare emergence and extinction timing using a Cox proportional-hazards model, a semi-parametric survival model that relates covariates to the event hazard rate (Table 7). Runs that experienced no emergence or extinction event by the end of 2M time steps were treated as censored. For extinction, both larger world size and compass availability substantially reduce the hazard: the extinction rate at 128×128 is approximately 25 times that at 512×512 ($HR=0.04$), and enabling the compass produces an effect of comparable magnitude ($HR=0.03$). For mining emergence, larger worlds develop mining substantially faster ($HR=6.07$), and compass availability accelerates emergence markedly ($HR=34.59$). The models exhibit strong discrimination between strata (concordance 0.80 for extinction, 0.85 for mining). We interpret these results as follows: larger worlds sustain larger populations, which lowers the probability of reaching the absorbing zero-population boundary while simultaneously affording exploration of a larger phenotype space through a greater aggregate mutation rate than in smaller worlds.

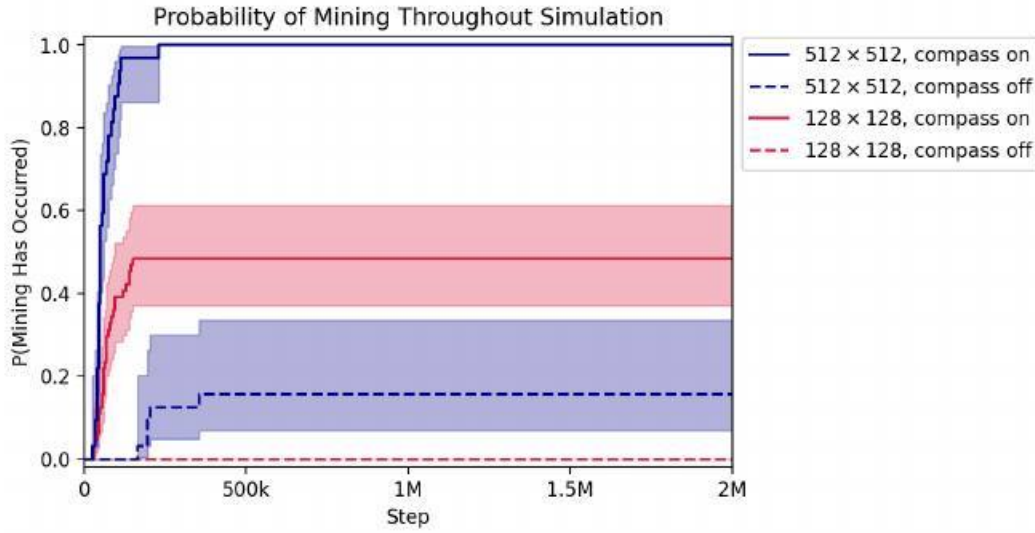


Figure 8: Estimated probability that mining has evolved within the population, conditioned on world size, compass access, and timestep. A separate non-parametric Kaplan–Meier estimator is fit for each stratum of runs (e.g., 512×512 , compass on). Shaded regions denote the 95% confidence interval for each curve. Each 512×512 stratum is fit on 32 runs and each 128×128 stratum on 64 runs.

Table 7: Cox proportional-hazards models for extinction and emergence events, using the 128×128 , compass-off condition as the baseline. A hazard ratio (HR) below 1 indicates a reduced hazard, i.e., a longer expected time to the event; for example, extinction $HR = 0.04$ for 512×512 indicates that extinction occurs at 0.04 times the rate observed at 128×128 . The analysis comprises 192 runs in total: 32 seeds with compass on and 32 with compass off at 512×512 , and 64 seeds with compass on and 64 with compass off at 128×128 .

Covariate	Extinction			Mining		
	HR	95% CI	<i>p</i> -value	HR	95% CI	<i>p</i> -value
Size 512×512	0.04	0.02–0.07	< 0.005	6.07	3.61–10.20	< 0.005
Compass On	0.03	0.01–0.05	< 0.005	34.59	13.51–88.57	< 0.005
Concordance	0.80			0.85		
Events / Total	135 / 192			68 / 192		

C ADDITIONAL FORAGING AND PREDATION RESULTS

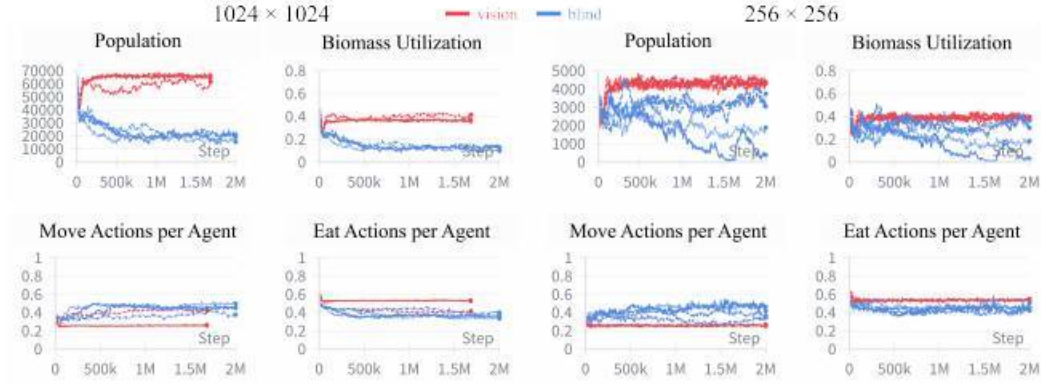


Figure 9: Foraging behavior in the **Ocean** world across scales. Enabling vision (red) yields more stable populations and greater biomass utilization, attributable to more frequent and more effective eating and to reduced movement. At larger scales these dynamics are more stable, with lower inter-run variance. Due to a constraint on available GPU hours, the vision seeds at the 1024×1024 resolution were run for 1.7M rather than 2M steps.

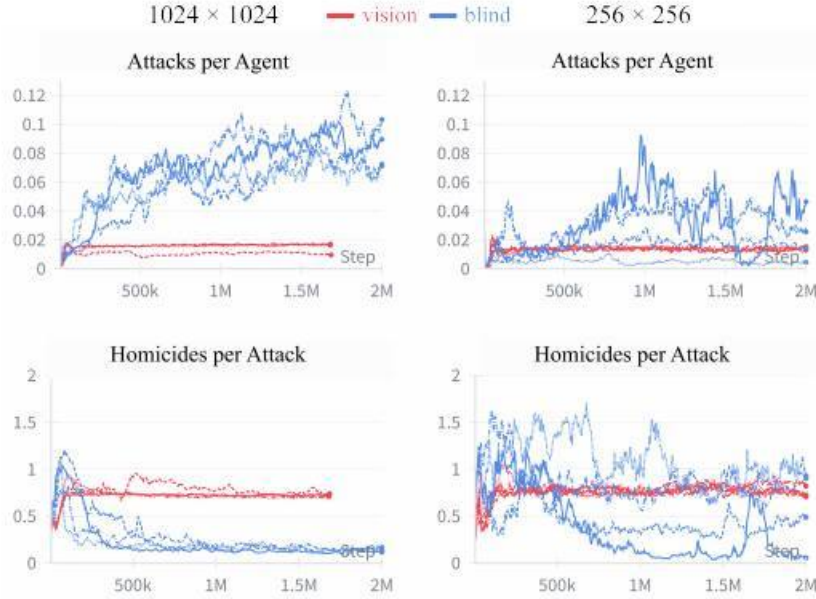


Figure 10: Predation behavior in the **Ocean** world across scales. With vision (red), agents attack less frequently but with substantially higher precision, consistent with the emergence of specialized predation behavior in addition to the foraging capability of Figure 9. This effect is again more pronounced at larger scales, with greater variance in the behavior of blind agents (blue) at smaller world sizes.

D ADDITIONAL EXPERIMENT VISUALIZATIONS

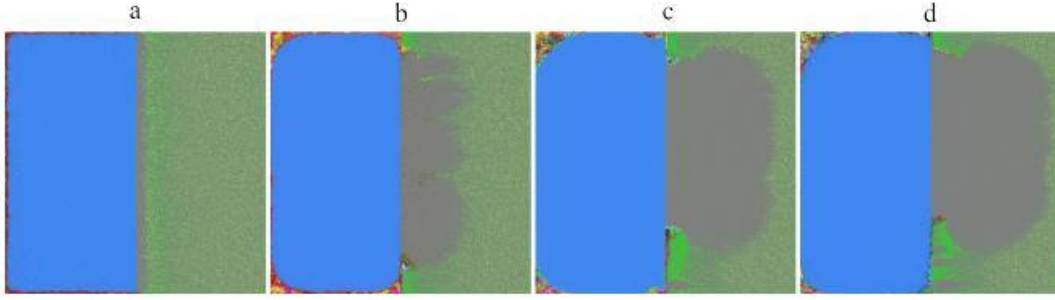


Figure 11: Larger-scale environments induce more varied and complex ecologies that give rise to distinctive spatial patterns. In this 1024×1024 **Beach** world, **(RC)** agents initially died at the shore (a), transferring biomass to dry land. They subsequently evolved mining behavior, clearing biomass from the beach (b). When biomass re-accumulated at the ends of the beach (c), subpopulations independently re-evolved mining behavior, traversing inland along these secondary biomass fronts to extract the resources (d).

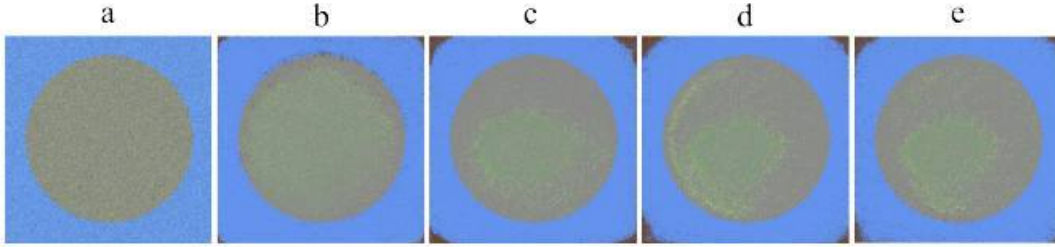


Figure 12: Mining behavior emerges across varied terrains. In this 512×512 **Island** world with **(RC)** agents, distinct regions of the shore are mined over the course of the run (b) (c). Biomass is transferred back onto parts of the island once mining becomes difficult and agents die before returning to water (d). Agents subsequently adapt to re-mine this accumulated biomass front, clearing it from the shore (e).

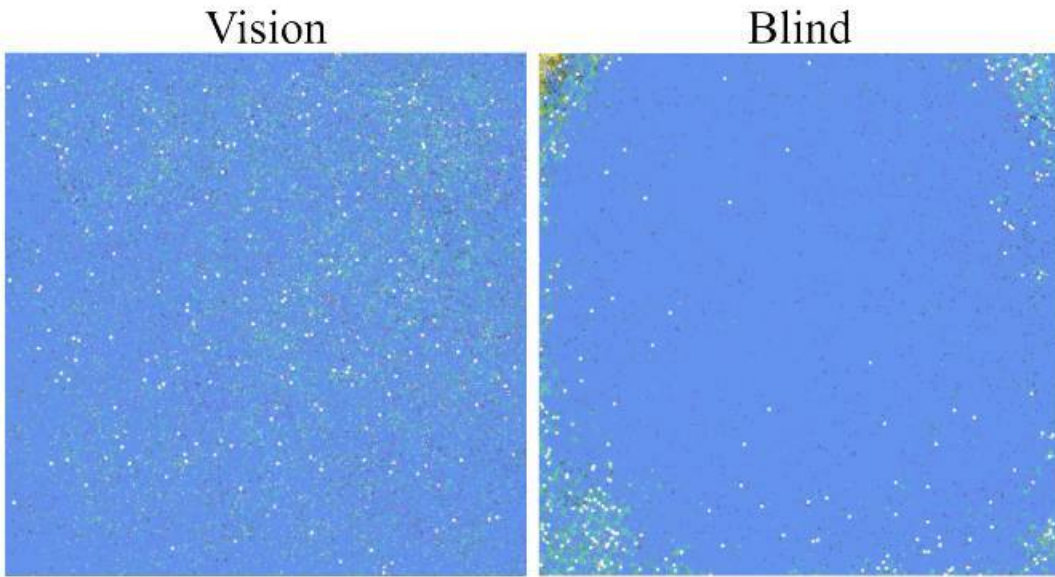


Figure 13: Representative runs in a 512×512 **Ocean** world with the attack action enabled: vision **(RCV+A)** agents (left) and blind **(RC+A)** agents (right). White squares mark cells at which an agent is attacking. With vision, agents adapt to locate free resources throughout the ocean and to prey on other agents more effectively (cf. Figures 5 and 6). Blind agents explore the ocean for resources to a far lesser extent, instead adopting policies that cluster in the corners of the world.

E DETAILS ON COMPUTATION AND REPRODUCIBILITY

This appendix details the simulator design, runtime measurements, and reproducibility considerations.

Simulator design. All experiments are conducted using a JAX-based, GPU-accelerated simulator designed for large-scale multi-agent ecosystems with open-ended populations. The simulator executes every component of the environment step on the GPU—terrain dynamics, resource flows, agent updates, and measurement—so that policy evaluation and simulation are compiled within a single JAX pipeline. The simulator exposes a minimal population-level interface: given the current map state, agent states, and heritable traits, a population algorithm returns actions and updated traits and resolves births, deaths, and environmental transitions. This separation of concerns permits the same environment implementation to be reused across distinct evolutionary algorithms, reinforcement learning methods, or other population update rules. All operations are implemented as pure JAX functions and combined with vectorization and just-in-time compilation, enabling fusion into a small number of GPU kernels. This design attains the throughput required for our scaling experiments; runtime measurements on a single H100 GPU are reported in Table 8.

A central feature of the simulator is its integrated tooling. A measurement API computes and records aggregate statistics at every step—including biomass utilization, movement and eating rates, attack and homicide rates, and an extensive set of environment-level parameters—which we consume directly in our figures and tables. In addition, an interactive 3D viewer renders terrain, water, resources, and agents from the generated reports and supports inspection of individual agents (Figure 1). This tooling was essential for discovering and validating qualitative phenomena, such as long-range resource mining and predation patterns, that are not apparent from scalar metrics alone.

From a high-performance-computing standpoint, the simulator addresses a gap in the existing landscape. General-purpose agent-based frameworks such as MASON (Luke et al. 2005), FLAME-GPU (Richmond et al. 2024), and Griddly (Bamford et al. 2020) offer high configurability but require substantial environment engineering and do not directly target open-ended, evolving populations at the scales we study. Conversely, GPU-native multi-agent RL environments are typically specialized to a fixed number of agents and task-specific reward structures. Our simulator combines GPU-native execution, explicit support for births and deaths with mutable traits, and configurable resource and terrain dynamics, thereby enabling open-ended populations exceeding 10^5 agents while remaining interoperable with a range of population algorithms. This combination constitutes a state-of-the-art platform for scaling ecological experiments of the kind reported here and a foundation for future work on emergent behavior in large populations. Our simulator will be released upon acceptance.

Runtime data. Table 8 reports runtime measurements across grid resolutions and the corresponding maximum agent populations. Each run was executed on a single H100 GPU for a maximum of 2M steps. Although the table reports runtimes for (RC) and (RCV + A) agents with 2 MLP layers and 64 channels per layer ($\sim 22k$ parameters total), the measured runtime depends on the agent network size, the active sensor set, and the logging verbosity.

Our largest runs, at the 1024×1024 resolution, require approximately 10 hours in the **Beach** terrain and approximately 14 hours in the **Ocean** terrain on a single H100, with peak memory usage of approximately 60GB. **Ocean** runs are marginally slower due to the enabled attack action and vision sensing.

Reproducibility. Owing to the large scale and stochastic initialization of our environments, the precise outcome of an experiment can vary substantially across runs. We therefore report results over four distinct random seeds for each experiment. Moreover, as a consequence of nondeterminism in certain low-level GPU operations, repeating an experiment with an identical random seed does not reliably reproduce the same trajectory. Specifically, modern GPUs may execute parallel floating-point operations such as summations in varying orders across runs and may fuse operations in ways that alter rounding behavior. The resulting small perturbations from finite-precision arithmetic can accumulate over many steps into appreciable divergence of the simulation trajectory. The 16-bit weight representation can coarsen these rounding effects but is not their primary source. Despite this run-to-run variation, consistent high-level patterns emerge when results are aggregated over multiple seeds.

Table 8: Runtime measurements in the **Beach** and **Ocean** worlds across grid resolutions and maximum agent populations. The **(RC)** agents each use a 22,465-parameter network and the **(RCV + A)** agents each use a 22,529-parameter network. We report simulation throughput in steps per second (hz) and the extrapolated total wall-clock runtime in hours for 2M steps. Effective runtime is lower for seeds whose populations go extinct before 2M steps.

	1024 × 1024		512 × 512		256 × 256		128 × 128		64 × 64	
	131072		32768		16384		8192		4096	
Environment	hz	runtime	hz	runtime	hz	runtime	hz	runtime	hz	runtime
Beach (RC)	55	10.1 hrs	190	2.9 hrs	315	1.8 hrs	535	1.0 hrs	750	0.7 hrs
Ocean (RCV + A)	40	13.9 hrs	142	3.9 hrs	235	2.4 hrs	415	1.3 hrs	750	0.7 hrs

F ABLATIONS

We present a series of ablations that quantify the effect of individual hyperparameters and assess the robustness of our results under varying environment and agent configurations.

F.1 INITIAL POPULATION SIZE SWEEP FOR FIXED MAP SIZE

We evaluate the effect of the initial population size in a 256×256 **Beach** world on the consistency of mining behavior and on population stability. The reported results are for **(RC)** agent populations. Agents without a compass (**R**) exhibited a 0% mining rate and a 100% extinction rate across all initial population sizes except the largest tested (8192), for which the mining and extinction frequencies were 25% and 75%, respectively. Note that the mining and extinction frequencies need not sum to 100%: a population may, for instance, develop mining and subsequently go extinct, or persist for the full 2M steps without ever developing mining.

As shown in Figure 14 the initial population size has no significant effect on either mining consistency or the extinction rate: populations initialized with only 128 agents evolve mining as frequently as those initialized with 8192 agents, and the extinction rate is largely invariant to the initial population size.

Figure 15 offers a mechanistic account of this insensitivity. Independent of the initial population size, every run converges rapidly to a comparable population size within the first 50,000 steps. This behavior is consistent with the environment implicitly defining a carrying capacity determined by the fixed resources in the system: populations initialized below the carrying capacity grow toward it, while populations initialized above it are driven down to it. Consequently, the effective population size is comparable across conditions following an early transient correction, which leaves the mining and extinction frequencies unaffected. Larger initial populations do contribute marginally to the total system resources, since each agent carries biomass that is deposited on death; accordingly, larger initial populations attain higher maximum populations on average over the run (Table 9), but this effect is too small to alter the mining or extinction frequency in this setting. We note that although the population size converges early, it does not remain constant thereafter: later in the run, populations across seeds can diverge substantially, as the ecological consequences of agents evolving distinct capabilities at distinct times reshape the subsequent resource distribution.

F.2 VARYING RESOURCE RULES

We evaluate the effect of two resource hyperparameters—the biomass photosynthesis rate and the initial biomass site density—on the mining and extinction frequencies. The experimental setting matches that of Appendix F.1. The initial population size is fixed at 2048 for this study, and agents again have a compass (**RC**).

The biomass photosynthesis rate governs the rate at which landscape biomass generates energy via photosynthesis, thereby supplying resources to agents over time. The initial biomass site density parameter specifies the fraction of grid cells containing biomass at initialization. All experiments

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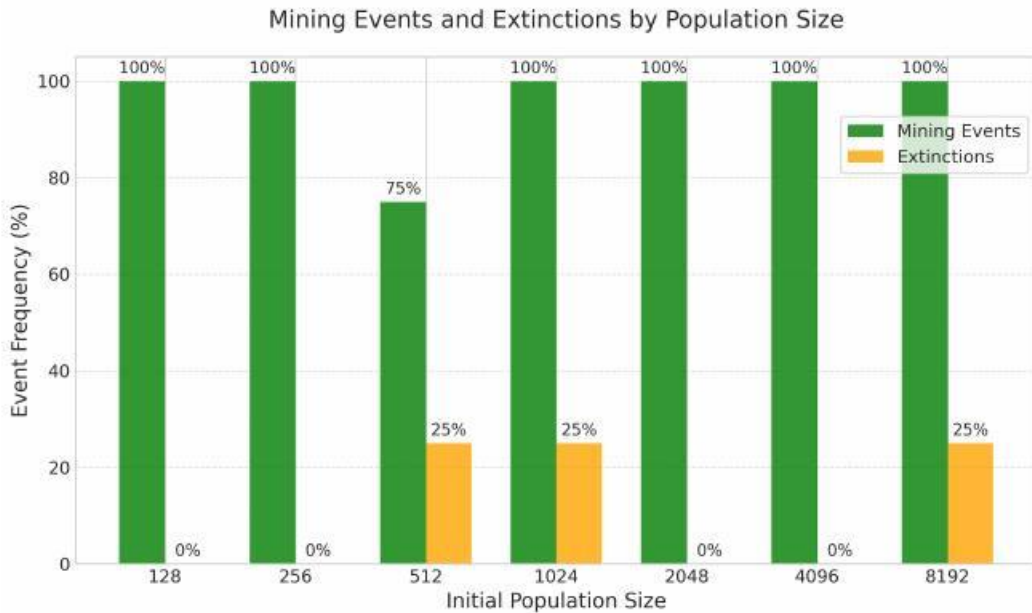


Figure 14: Frequency of mining and extinction events in a 256×256 **Beach** world across initial (**RC**) agent population sizes. Each frequency is computed over four random seeds per initial population size; e.g., a 25% extinction-event frequency indicates that the population reached zero before 2M steps in 1 of the 4 seeds.

Table 9: Maximum population size attained after step 0 as a function of the initial population size, reported as the mean and standard deviation over four random seeds. Larger initial population sizes attain marginally higher maximum populations, with greater inter-seed variance, than smaller initial population sizes. The attained maximum is nonetheless largely robust to the initial population size: an initial size of 128—only 1.6% of an initial size of 8192—attains 78.9% of the maximum population attained by the latter (3178 vs. 4026).

Initial Population Size	Max Population Size After Step 0
128	3178 ± 217
256	3056 ± 61
512	3451 ± 437
1024	3613 ± 381
2048	3786 ± 431
4096	3689 ± 437
8192	4026 ± 642

in the main body (Section 4) use a biomass photosynthesis rate of 0.01 and an initial biomass site density of $1/4$.

Whereas varying the initial population size did not affect the mining and extinction frequencies (Figure 14), varying the resource rules has a pronounced effect, as demonstrated in Figures 16 and 17

Mechanistically, increasing either the biomass photosynthesis rate or the initial biomass site density directly raises the effective carrying capacity of the environment by increasing the total resources in the system. Larger populations are thereby sustained throughout the run, which increases the probability of mutations favoring mining behavior while decreasing the probability of extinction. Our measurements support this account: averaging over seeds that did not go extinct, runs with

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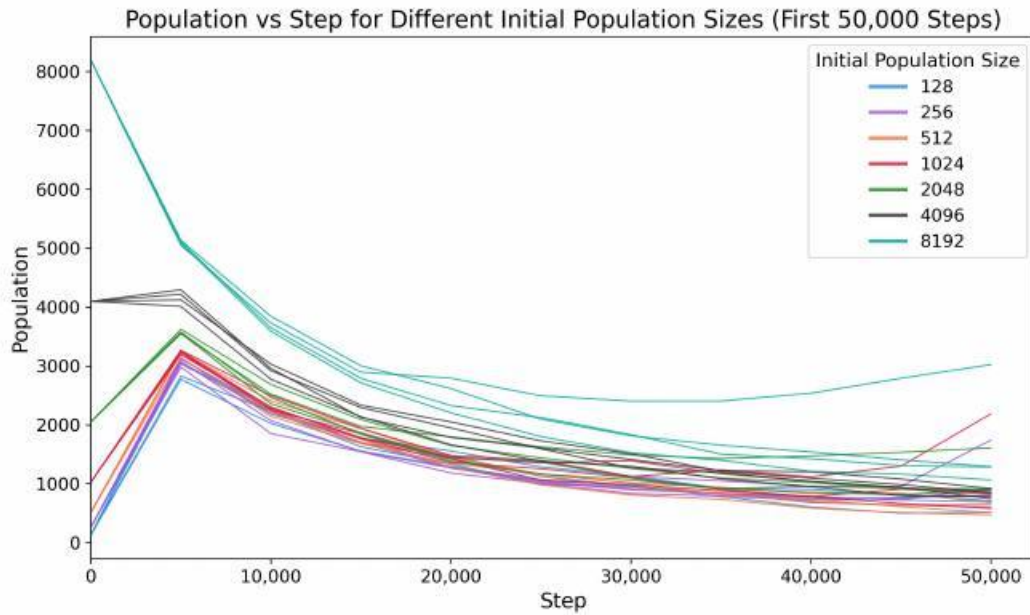


Figure 15: Population size over the first 50,000 time steps in the 256×256 **Beach** with (RC) agents, with four seeds per initial population size. The environment supports an implicit carrying capacity determined by the resources available in the system, and all initial population sizes converge toward this value early in the run. Larger initial population sizes marginally increase the total system resources through biomass deposited on agent death, raising the carrying capacity slightly.

initial biomass site density $1/16$ attained a maximum population of 1608 after step 0, versus 2466 at density $1/8$, 3786 at density $1/4$, and 5353 at density $1/2$.

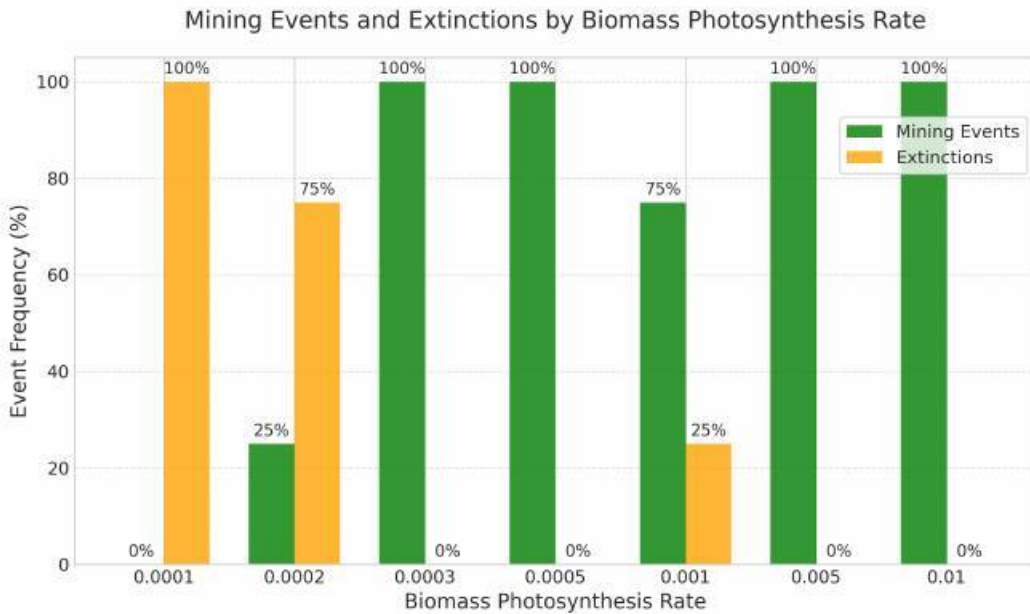


Figure 16: Frequency of mining and extinction events in a 256×256 **Beach** world across settings of the biomass photosynthesis rate. Each frequency is computed over four random seeds per setting. Mining emerges consistently at biomass photosynthesis rates of 0.0003 and above.

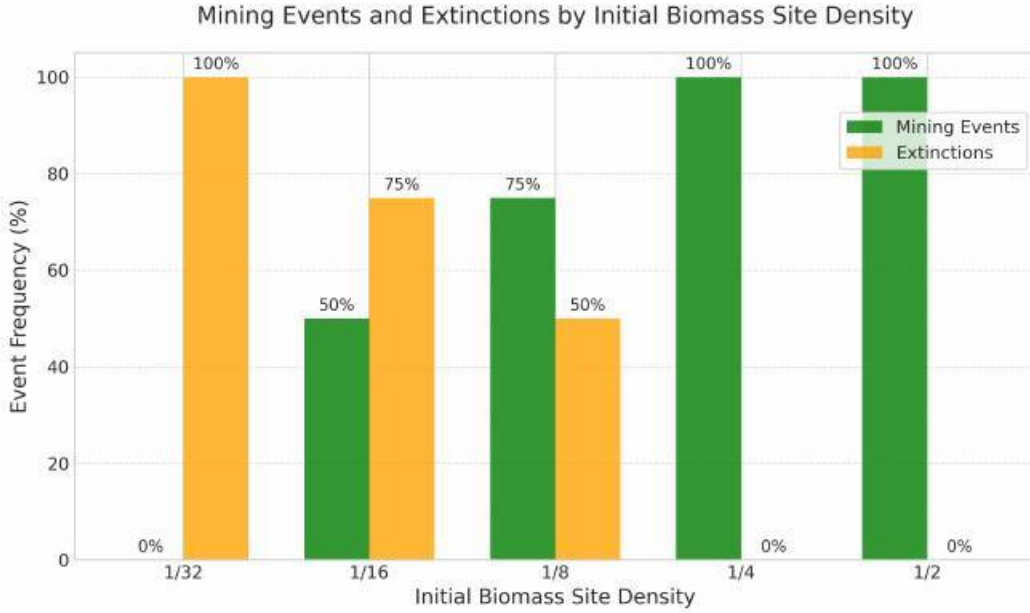


Figure 17: Frequency of mining and extinction events in a 256×256 **Beach** world across settings of the initial biomass site density. Each frequency is computed over four random seeds per setting. Higher initial biomass site densities yield more consistent discovery of mining and a lower probability of extinction.

F.3 RULING OUT SELF-AVERAGING IN LARGER POPULATIONS

One potential concern is that the increased stability of mining behavior at large scale (Table 1) is attributable purely to statistical self-averaging rather than to qualitatively distinct dynamics. By self-averaging we invoke the statistical-physics principle that, as the system size N grows, fluctuations in macroscopic quantities decay approximately as $1/\sqrt{N}$, so that a single large system behaves like the ensemble average over smaller systems of the same aggregate size. Under this hypothesis, a 512×512 **Beach** world would behave as an ensemble of many weakly coupled 128×128 worlds evaluated in parallel, and our scaling results could in principle be recovered by running a sufficient number of small worlds and averaging over random seeds.

To test this hypothesis, we conduct an experiment that holds the total “sample size” approximately fixed while varying its partition across environments. In the **Beach** terrain with compass-equipped agents (**RC**), we consider three conditions:

- 4 runs of a 512×512 world with 8192 initial agents each,
- 16 runs of a 256×256 world with 2048 initial agents each,
- 64 runs of a 128×128 world with 512 initial agents each.

In each case, the total number of grid cells across runs, the total starting resources and the total initial number of initial agents are identical:

$$4 \times 512^2 = 16 \times 256^2 = 64 \times 128^2, \quad 4 \times 8192 = 16 \times 2048 = 64 \times 512.$$

Each seed is run for 2M steps under the same hyperparameters as in Section 4.1. As a control, we repeat the protocol for agents without a compass (**R**). Table 10 summarizes the long-distance mining and extinction frequencies for each configuration.

If large environments behaved as self-averages of many smaller ones, the mining and extinction rates would be approximately invariant to the partition of grid cells and agents across runs, and 4 large worlds and 64 small worlds would exhibit comparable mining probabilities. We instead

observe a strong dependence on world size. For **(RC)** agents, mining emerges in all runs with no extinctions at 512×512 , remains common but not universal at 256×256 , and becomes unreliable and fragile at 128×128 , where many populations go extinct before 2M steps. In contrast, **(R)** agents never develop mining and go extinct at all scales, confirming that the compass remains a necessary enabler of long-range behavior.

Table 10: Mining and extinction events under a trade-off between world size and the number of independent runs. We compare **Beach** worlds at grid resolutions 512×512 , 256×256 , and 128×128 with initial populations scaled proportionally (8192, 2048, and 512 agents, respectively). For each resolution we hold the aggregate number of grid cells and initial agents fixed across runs by using 4 seeds at 512×512 , 16 seeds at 256×256 , and 64 seeds at 128×128 . We report the fraction of runs in which mining emerged () and the fraction in which the population went extinct before 2M steps () for agents with a compass (**(RC)**) and without a compass (**(R)**).

	512 × 512 8192		256 × 256 2048		128 × 128 512	
						
(RC)	4/4	0/4	15/16	1/16	31/64	41/64
(R)	0/4	4/4	0/16	16/16	0/64	64/64

These results are inconsistent with a self-averaging explanation. Rather, larger environments appear to enable and stabilize mining behavior in a manner that cannot be reproduced by averaging over a larger number of small simulations.

Even with the total number of cells, agents, and simulation steps matched across conditions, concentrating resources into a few large worlds yields a regime in which mining is consistent and extinctions are rare, whereas dispersing the same resources across many small worlds yields populations that frequently fail to discover mining and are more prone to collapse. This indicates that increasing environment scale does not merely suppress stochastic fluctuations but modifies the underlying ecological dynamics—for example, by sustaining larger, more spatially distributed populations and more persistent diversity in the evolving policy set.

F.4 MUTATION RATE SWEEP

We assess the sensitivity of mining emergence to the mutation-rate hyperparameter, which sets the standard deviation of the Gaussian noise added to an offspring’s policy at birth and therefore controls the expected divergence between an offspring and its parent. We again run experiments in a 256×256 **Beach** world with an initial population of 2048 **(RC)** agents for a maximum of 2M steps.

Figure 18 shows that, for mutation rates between 6×10^{-3} and 7.875×10^{-2} , mining emerges in at least 3 of the 4 seeds, indicating that our main results (Section 4.1) at mutation rate 3×10^{-2} are robust throughout this interval. As expected, mutation rates that are too small (e.g., 3×10^{-3}) prevent agents from developing new behavior and impair survival, resulting in extinction, whereas mutation rates that are too large (e.g., 3×10^{-1}) destabilize the population, consistently resulting in extinction without mining adaptation.

F.5 NETWORK SIZE

We study the effect of the agent policy network’s size and architecture on the mining and extinction frequencies, together with the modulating effect of environment scale. We consider 256×256 and 512×512 **Beach** worlds with initial populations of 2048 and 8192 **(RC)** agents, respectively (scaling by a factor of four with the number of grid cells). We ran **(R)** baselines for each experiment, again observing rare-to-no mining events and 100% extinction rates.

The main experiments in Section 4 used agent policies with 2 layers and 64 channels, corresponding to 22465 parameters per agent (Table 11). Figure 19 shows that, in the 256×256 **Beach**, halving the channel count to 32 (9185 parameters) or doubling it to 128 (61313 parameters) at fixed depth

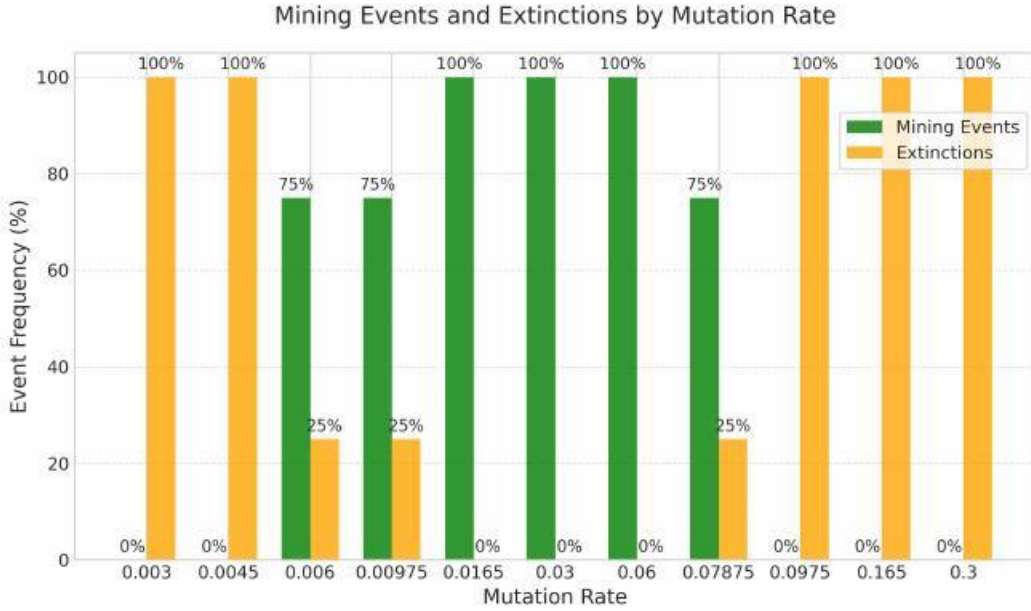


Figure 18: Frequency of mining and extinction events in a 256×256 **Beach** world across mutation rates. Each frequency is computed over four random seeds per setting.

has minimal-to-no effect on the mining and extinction frequencies. Varying the network depth, in contrast, has a pronounced effect: all populations with 1 layer and 32 channels went extinct, as did 50% of populations with 3 layers and 128 channels.

Critically, scaling the environment to the 512×512 resolution produced qualitatively different dynamics across network depths (Figure 20). In the larger world, populations with 3-layer, 128-channel networks mined and survived through 2M steps consistently, and populations with the small 1-layer, 32-channel networks likewise mined consistently. We hypothesize that both smaller and larger networks impede the discovery of new behavior, but for distinct reasons: a smaller network has insufficient capacity to represent good solutions, whereas a larger network induces a larger search space that lengthens the time required to locate promising behavior. Independent of the specific mechanism, the observation that these networks fail frequently in small environments but succeed in larger ones further illustrates the stabilizing effect of increased scale. A larger environment sustains larger populations, which affords greater exploration of policy space; concurrently, the longer time required for locally successful mutants to propagate across a large environment promotes diversity and thereby further increases exploration. For additional discussion of these effects, see (Mitchell et al. 2006).

Table 11: Total parameter count of (RC) and (R) agent policies across network architectures.

Architecture (layers, channels)	Parameter Count	
	(RC)	(R)
1, 32	8129	7969
2, 32	9185	9025
2, 64	22465	22145
2, 128	61313	60673
3, 128	77825	77185

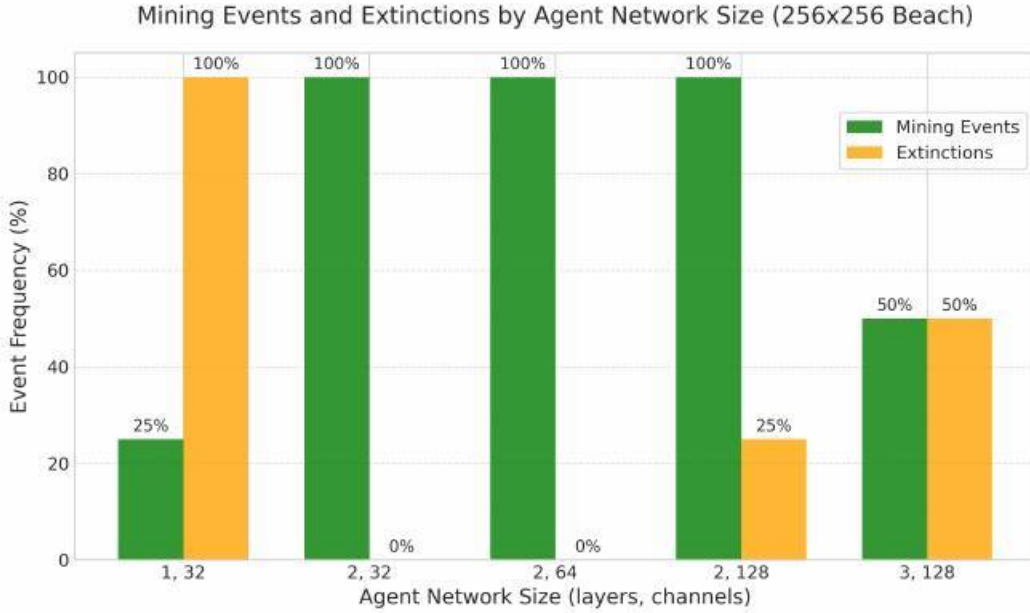


Figure 19: Frequency of mining and extinction events in a 256×256 **Beach** world across agent policy network sizes. Each frequency is computed over four random seeds per setting.

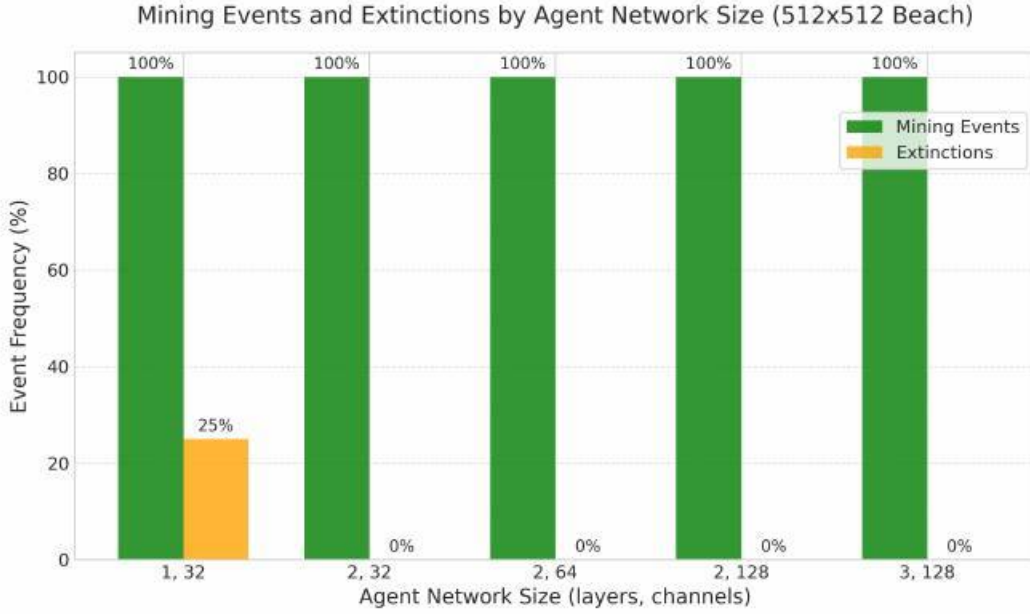


Figure 20: Frequency of mining and extinction events in a 512×512 **Beach** world across agent policy network sizes. Each frequency is computed over four random seeds per setting.

F.6 VARYING VISION RANGE

We assess the effect of the vision range and viewing configuration on predation and foraging behavior in the 512×512 **Ocean** world with (RCV+A) agents. We evaluate vision-sensor ranges of 3×3 , 7×7 , and 9×9 . For each range we evaluate the two viewing configurations shown in Figure 21 (a) Around, in which the agent perceives cells in all directions, and (b) Front, in which the agent does

not perceive cells behind it. The total number of cells in the agent’s field of view is held equal across the two configurations. The main vision experiments in Section 4.2 used the 7×7 Around setting.

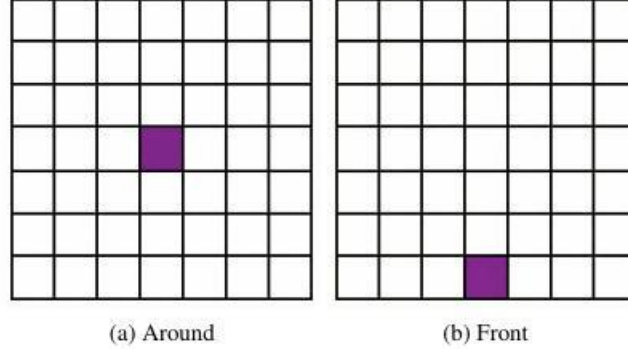


Figure 21: Comparison of the Around and Front viewing configurations at the 7×7 vision range. The purple cell denotes the agent, oriented toward the top of the page.

Table 12: Foraging and predation measurements in the 512×512 **Ocean** world for vision agents (**RCV+A**) across vision ranges and viewing configurations. Definitions of the metrics denoted by each icon are given in Tables 5 and 6. For each seed, each metric is averaged over all steps from step 500k onward to allow the metrics to stabilize. Reported values are the mean over the four seeds together with the standard deviation of the four per-seed averages.

Vision Range	Viewing Configuration						
3×3	Blind	7212 ± 2496	$.181 \pm .060$	$.435 \pm .090$	$.392 \pm .033$	$.032 \pm .006$	$.307 \pm .090$
	Around	10308 ± 4016	$.258 \pm .105$	$.352 \pm .014$	$.427 \pm .022$	$.040 \pm .030$	$.388 \pm .195$
	Front	13863 ± 520	$.351 \pm .023$	$.336 \pm .008$	$.437 \pm .009$	$.026 \pm .006$	$.470 \pm .145$
7×7	Around	16739 ± 122	$.373 \pm .003$	$.259 \pm .000$	$.532 \pm .001$	$.015 \pm .000$	$.751 \pm .007$
	Front	15255 ± 623	$.343 \pm .015$	$.253 \pm .001$	$.540 \pm .004$	$.015 \pm .000$	$.690 \pm .041$
9×9	Around	16744 ± 117	$.372 \pm .002$	$.257 \pm .001$	$.534 \pm .001$	$.015 \pm .000$	$.728 \pm .009$
	Front	15211 ± 1416	$.342 \pm .033$	$.253 \pm .001$	$.541 \pm .007$	$.015 \pm .000$	$.664 \pm .050$

These results indicate that the smaller 3×3 vision range yields behavior largely indistinguishable from no vision (Blind), while the 9×9 range yields behavior comparable to 7×7 . This is consistent with diminishing returns to larger vision ranges under the present environment configuration. Such a saturation is expected: because the agents are memoryless, it is unlikely that visual features at the periphery of a 9×9 window alter the optimal action relative to a 7×7 window consistently enough to provide a strong learning signal.

The measurements further suggest that, at the 7×7 and 9×9 vision ranges, the Around configuration is marginally more favorable in terms of population stability and the effectiveness of the attack and eat actions, yielding higher average population , biomass utilization , and homicides per attack  than the Front configuration. We attribute this to diminishing returns from perceiving more distant cells: under the Around configuration a greater fraction of the perceived cells are proximal and thus more likely to carry information relevant to near-term actions than the distal cells perceived under the Front configuration. At the 3×3 vision range, however, the ordering reverses, with the Front configuration outperforming the Around configuration; this indicates that, in this environment, the marginal value of perceiving one additional cell ahead in the 3×3 Front configuration exceeds that of perceiving one cell behind in the 3×3 Around configuration, the latter yielding behavior most similar to the Blind setting.

F.7 VARYING ATTACK STRENGTH

We study the effect of reducing the agent attack strength on emergent behavior and population dynamics in the 256×256 **Ocean** world. We evaluate three attack strengths: (1) Full Attack, dealing

10 HP of damage and thus instantly killing any agent within the attack radius; (2) Half Attack, dealing 5 HP of damage; and (3) Quarter Attack, dealing 2.5 HP of damage. The main **Ocean** experiments in Section 4.2 used the Full Attack setting.

Reducing the attack strength from Full Attack to Half Attack substantially reduces both the effectiveness and the frequency of attacks: attacks per agent and homicides per attack both decrease considerably (Figure 22). Consequently, populations under lower attack strength grow larger, with agents allocating more actions to movement and eating and thereby achieving greater biomass utilization. Further reducing the strength to the Quarter Attack setting yields diminishing returns, with little additional reduction in violence or additional population growth. Mechanistically, when each attack deals less damage, a homicide requires more attacks per agent, reducing the per-attack lethality; the reduced lethality in turn renders attacking less useful and therefore less strongly selected for in the population. Agents under lower attack strength consequently do not become as proficient at attacking and allocate more time to foraging.

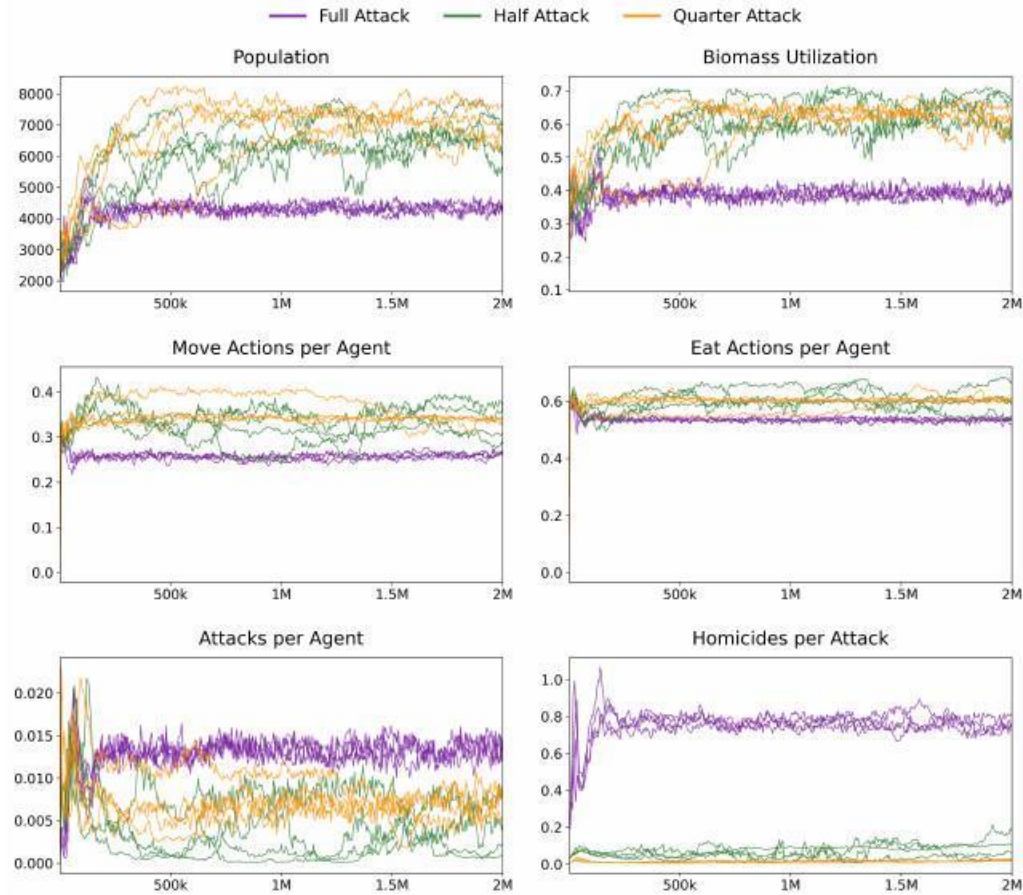


Figure 22: Population size and action-frequency measurements over 2M steps in a 256×256 **Ocean** world for (RCV+A) agents across the Full Attack, Half Attack, and Quarter Attack settings. Four seeds are plotted per attack-strength setting.

F.8 VARYING AGENT HP

We investigate the effect of reducing the starting HP of offspring agents and lowering the healing rate, thereby imposing selective pressure for agents to ‘grow’ over their lifetime in order to survive. The healing-rate parameter specifies the HP regenerated per healing step, which converts available energy into HP; an agent can influence its healing only indirectly, by maintaining sufficient energy.

We study two settings: (1) Full HP, in which offspring HP starts at 5 and the healing rate is 1 HP per 0.1 energy; and (2) Low HP, in which offspring HP starts at 1 and the healing rate is 0.5. The maximum agent HP is 10 in all settings. The main experiments (Section 4) used the Full HP setting.

Across the two settings, there is no significant difference in homicides per attack, eat-action frequency, or biomass utilization (Figure 23). Under Full HP, populations grow marginally larger; under Low HP, agents attack less frequently and move marginally more. We hypothesize that agents born with lower HP must prioritize foraging early in life in order to reach full size, which would account for this difference.

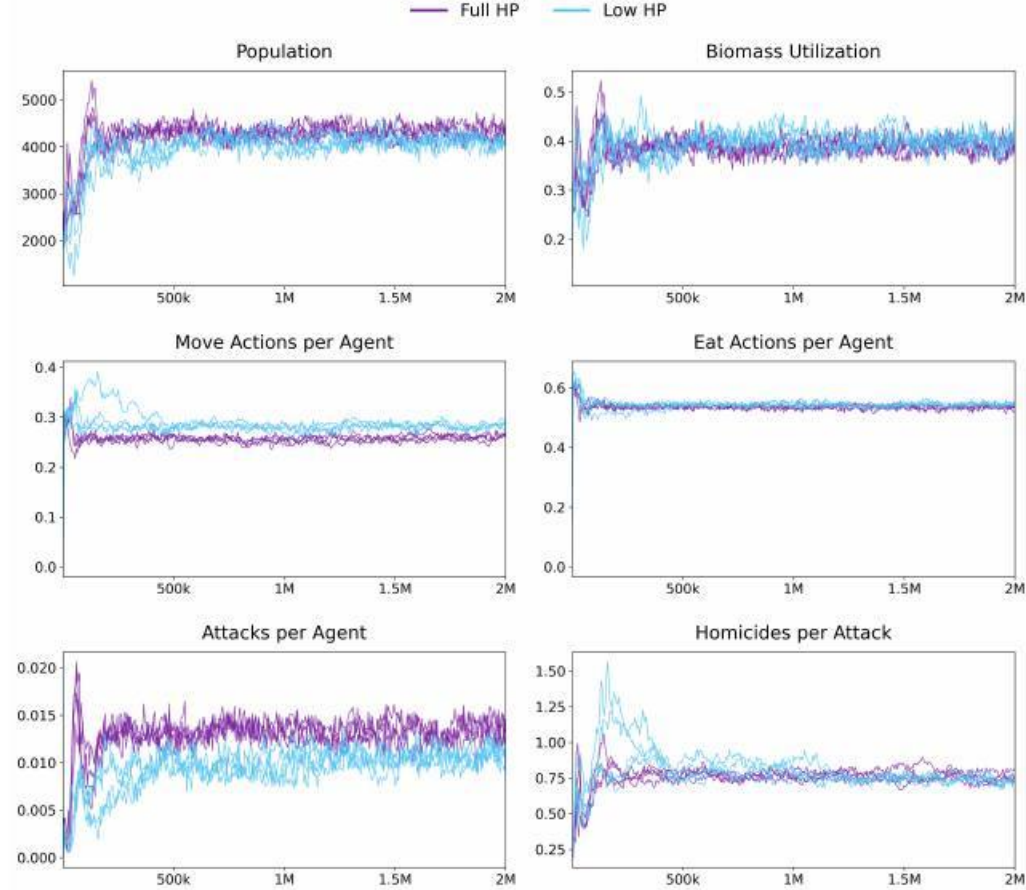


Figure 23: Population size and action-frequency measurements over 2M steps in the 256×256 Ocean world for (RCV+A) agents across the Full HP and Low HP settings. Four seeds are plotted per HP setting.